

CODE GOLF COMPETITION
100 marks

Q3
CODE GOLF COMPETITION
100 marks

Q4
CODE GOLF COMPETITION
100 marks

0/4 answered

Your Code

```
1 #include <stdio.h>
2 void quicksort(int a[], int low, int high) {
3     if (low < high) {
4         int i = low, j = high, pivot = a[low], temp;
5         while (i < j) {
6             while (a[i] <= pivot && i < high)
7                 i++;
8             if (a[j] > pivot)
9                 j--;
10
11             if (i < j) {
12                 temp = a[i];
13                 a[i] = a[j];
14                 a[j] = temp;
15             }
16         }
17         a[low] = a[j];
18         a[j] = pivot;
19         quicksort(a, low, j - 1);
20         quicksort(a, j + 1, high);
21     }
22 }
23 int main() {
24     int a[] = {50, 23, 9, 18, 61, 32};
25     int n = 6;
26     quicksort(a, 0, n - 1);
27     for (int i = 0; i < n; i++)
28         printf("%d ", a[i]);
29     return 0;
30 }
```

Run

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Input

stdin input

Output

9 18 23 32 50 61

QUESTIONS

Q1
CODE GOLF COMPETITION
100 marks

Q2
CODE GOLF COMPETITION
100 marks

Q3
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100 marks

Q4
CODE GOLF COMPETITION
100 marks

1/4 answered

Q2 CODE GOLF COMPETITION

100 marks

Write a program that implements the bubble sort

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int a[] = {50,23,9,18,61,31};
4     int n = 6, temp;
5     for (int i = 0; i < n-1; i++)
6         for (int j = 0; j < n-i-1; j++)
7             if (a[j] > a[j+1]) {
8                 temp = a[j];
9                 a[j] = a[j+1];
10                a[j+1] = temp;
11            }
12     for (int i = 0; i < n; i++)
13         printf("%d ", a[i]);
14     return 0;
15 }
```

Run

Save

Submit

Input

stdin input

Output

9 18 23 31 50 61

QUESTIONS

Q1
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100 marks

Q2
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Q3
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Q4
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2/4 answered

Q3 CODE GOLF COMPETITION

100 marks

Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int a[] = {9,5,1,4,3};
4     int n = 5, key, j;
5     for (int i = 1; i < n; i++) {
6         key = a[i];
7         j = i - 1;
8         while (j >= 0 && a[j] > key) {
9             a[j + 1] = a[j];
10            j--;
11        }
12        a[j + 1] = key;
13    }
14    for (int i=0; i<n; i++)
15        printf("%d ", a[i]);
16    return 0;
17 }
```

Run

Save

Submit

Input

stdin input

Output

1 3 4 5 9

Q1
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Q2
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Q3
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100 marks

Q4
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100 marks

3/4 answered

Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10

Your Code

```
1 #include <stdio.h>
2 void merge(int a[], int l, int m, int r) {
3     int i=l, j=m+1, k=0, b[10];
4     while (i<=m && j<=r) b[k++] = a[i]<a[j]?a[i++]:a[j++];
5     while (i<=m) b[k++] = a[i++];
6     while (j<=r) b[k++] = a[j++];
7     for (i=l, k=0; i<=r; i++) a[i] = b[k++];
8 }
9 void sort(int a[], int l, int r) {
10    if (l<r) {
11        int m = (l+r)/2;
12        sort(a, l, m); sort(a, m+1, r);
13        merge(a, l, m, r);
14    }
15 }
16 int main() {
17    int a[] = {38, 27, 43, 3, 9, 82, 10}, n=7;
18    sort(a, 0, n-1);
19    for (int i=0; i<n; i++) printf("%d ", a[i]);
20 }
```

▶ Run

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🚀 Submit

Input

stdin input

Output

3 9 10 27 38 43 82